

Aaron Dutta

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EXECUTIVE SUMMARY

A highly dedicated and result-driven individual who is passionate about mathematics and programming, striving to harness skills in the field of engineering and software.

EDUCATION

University of Waterloo

Waterloo, ON

BMath, Honours Mathematics with Co-Op

Expected June 2031

EXPERIENCE

Mathematics Mentor

2024 – Present

Filix School of Education

- Mentoring several Grade 7-12 students through weekly math circles that encourage collaborative problem solving
- Advancing students' critical thinking skills by combining theoretical concepts with practical exercises

Swim Coach

2023 – Present

Olympian Swimming

- Delivering structured swim lessons focused on skill development, swim posture, and stroke technique
- Maintaining attendance and evaluating progress through comprehensive report cards to track growth

PROJECTS

FraudGen | *Python, Anthropic API, FastAPI, Streamlit, Pandas, NetworkX, Pydantic*

[GitHub](#)

- Designed a **4-agent adversarial LLM pipeline** that explores fraud variant space via a **coverage matrix** and generates **persona-seeded mule-network** scenarios
- Shipped **async parallel orchestration**, **deterministic validation/repair**, and **critic scoring** with retry loops and **ML-ready exports** for **XGBoost/GNN** training
- Built a **Streamlit** monitoring console with **network graph visualization** and dataset browser; exposed the pipeline through **FastAPI REST endpoints**

MyCelium | *Python, FastAPI, React, TypeScript, SciPy, Gemini API, Google Maps*

[GitHub](#)

- Built a **SciPy MILP optimizer** for **comparative-advantage crop assignment** using **gap-weighted yield matrices** and **hub reachability constraints**
- Shipped a **FastAPI** backend with **14+ dimension suitability scoring**, a **transaction-matching engine**, and **Gemini-powered** personalized grow-cycle task generation
- Delivered a **React/TypeScript** farmer and admin UI with onboarding, wallet, food marketplace, and **Google Maps network visualization**

TechniqueTitan | *Python, OpenCV, MediaPipe, NumPy*

[GitHub](#)

- Built a real-time **computer vision** prototype using **MediaPipe** and **OpenCV** to extract **21 hand landmarks** from live webcam input
- Implemented a modular **HandDetector** class with configurable **detection/tracking confidence** and **multi-hand support** for piano posture analysis
- Structured the pipeline for a future **feedback engine** with **heuristic scoring** to evaluate hand posture and technique in real time

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, HTML5/CSS3, SQL, Swift

Frameworks/Platforms: AppKit, Flask, React, SwiftUI

Technologies/Tools: Docker, Figma, Git, MediaPipe, OpenCV, PyCharm, RESTful APIs, SQLite

CERTIFICATIONS

- Harvard University's CS50x 2024
- Royal Conservatory of Music Level 9 Piano, Level 10 Music History 2023 – 2024
- Bronze Medallion, Bronze Cross, Emergency First Aid, Standard First Aid 2023